Flow Matching for Generative Modeling

From CNFs to Straight Flows and Beyond

Flow Matching Reading Group

March 20, 2026

Outline

1. Motivation & Background
2. Flow Matching: Core Theory
3. Rectified Flow & Straight Trajectories
4. Generalizations & Unifications
5. Extensions
6. Practical Considerations & Impact
7. Conclusion

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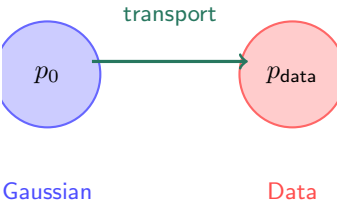
The Goal: Learning to Transport Distributions

Central problem in generative modeling:

- We have samples from $p_{\text{data}}(x)$ (e.g., images)
- We want to *generate* new samples from p_{data}
- Standard approach: learn a mapping from a simple noise distribution $p_0 = \mathcal{N}(0, I)$ to p_{data}

Desiderata:

- Tractable training
- Fast inference (few function evaluations)
- High sample quality
- Exact likelihood (bonus)



Continuous Normalizing Flows (CNFs)

Key Idea

Define a time-dependent vector field $v_\theta(x,t)$ that continuously transforms $p_0 \to p_{\text{data}}$ via an ODE:

$$\frac{dx}{dt} = v_\theta(x,t), \quad x(0) \sim p_0, \quad x(1) \approx p_{\text{data}}$$

Change of variables / likelihood:

$$\log p_1(x) = \log p_0(x_0) - \int_0^1 \nabla \cdot v_\theta \, dt$$

Training via maximum likelihood requires integrating the ODE *during every gradient step*.

The Simulation Bottleneck

- ODE solve per training step is **expensive**
- Adjoint method needed for gradients
- Unstable training dynamics
- Doesn't scale to ImageNet

Score-Based Diffusion Models

DDPM / Score SDE: Add noise forward, learn to reverse

Forward process (fixed):

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0,\ (1-\bar{\alpha}_t)I)$$

Training (denoising score matching):

$$\mathbb{E}\left\|\epsilon_\theta(x_t, t) - \epsilon\right\|^2$$

Strengths: simulation-free training, scalable
Weaknesses: many NFE at inference, fixed Gaussian noise schedule

Question

Can we get the **training simplicity** of diffusion *and* the **flexibility + efficiency** of CNFs?

Yes — this is Flow Matching.

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Flow Matching (Lipman et al., ICLR 2023)

"Flow Matching for Generative Modeling" — Lipman, Chen, Ben-Hamu, Nickel, Le (Meta AI)

The Key Insight

Instead of training by *maximizing likelihood* (which requires ODE simulation), directly **regress** v_θ onto a known target vector field u_t.

Flow Matching (FM) objective:

$$\mathcal{L}_{\mathsf{FM}}(\theta) = \mathbb{E}_{t,\, x\sim p_t} \left\| v_\theta(x,t) - u_t(x) \right\|^2$$

Problem

The marginal vector field $u_t(x)$ is **intractable** — it requires marginalizing over all data points $x_1 \sim p_{\mathsf{data}}$.

Conditional Flow Matching: Making it Tractable

Key observation: condition on a single data point x_1, define a simple path from p_0 to δ_{x_1}.

Conditional probability path:

$$p_t(x \mid x_1), \quad p_0(x \mid x_1) = \mathcal{N}(x; 0, I), \quad p_1(x \mid x_1) \approx \delta_{x_1}$$

Conditional FM (CFM) objective:

$$\boxed{\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\, x_1 \sim p_{\text{data}},\, x_t \sim p_t(\cdot \mid x_1)} \left\| v_\theta(x_t, t) - u_t(x_t \mid x_1) \right\|^2}$$

Theorem (Lipman et al., 2022)

$\nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$ *(same gradients — CFM is a valid proxy!)*

Each x_1 gives a simple tractable path. Marginalize at inference time via the learned v_θ.

Gaussian Conditional Paths

The simplest conditional path: a **Gaussian** centered on x_1, shrinking variance over time.

$$p_t(x \mid x_1) = \mathcal{N}\big(x;\ \mu_t(x_1),\ \sigma_t^2 I\big)$$

Sample at time t:

$$x_t = \mu_t(x_1) + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Conditional vector field (in closed form):

$$u_t(x \mid x_1) = \frac{\dot{\sigma}_t}{\sigma_t}(x - \mu_t(x_1)) + \dot{\mu}_t(x_1)$$

Diffusion path (recover DDPM):
$\mu_t = \sqrt{\bar{\alpha}_t} x_1$, $\sigma_t = \sqrt{1 - \bar{\alpha}_t}$

OT path (linear, Lipman et al.):
$\mu_t = t\, x_1$, $\sigma_t = 1 - (1 - \sigma_{\min})t$

The Optimal Transport Path

With $\mu_t(x_1) = t\,x_1$ and $\sigma_t = 1 - (1 - \sigma_{\min})t$ (small $\sigma_{\min}$):

Sample:

$$x_t = (1-t)\,x_0 + t\,x_1, \quad x_0 \sim \mathcal{N}(0, I)$$

Target velocity:

$$u_t(x_t \mid x_1) = x_1 - x_0 \quad \text{(constant! independent of } t\text{)}$$

Why this is great

- **Linear interpolation** between noise and data — the simplest possible path
- **Constant velocity field** per trajectory $\rightarrow$ neural net has an easy target
- Trajectories are **straighter** than diffusion paths $\rightarrow$ fewer NFEs at inference

Flow Matching Algorithm

Training

Repeat:

1. $x_1 \sim p_{\text{data}}$
2. $x_0 \sim \mathcal{N}(0, I)$
3. $t \sim \text{Uniform}(0, 1)$
4. $x_t = (1-t)x_0 + t\,x_1$
5. $\mathcal{L} = \|v_\theta(x_t, t) - (x_1 - x_0)\|^2$
6. Update θ via $\nabla_\theta \mathcal{L}$

Sampling

1. $x_0 \sim \mathcal{N}(0, I)$
2. Choose ODE solver (Euler, RK4, ...)
3. **For** $t = 0 \to 1$ with step h:
   $$x_{t+h} = x_t + h \cdot v_\theta(x_t, t)$$
4. **Return** x_1

No score function. No ELBO. Just regression.

Results: FM vs. Diffusion on ImageNet

Lipman et al. (2022): First CNF to scale to ImageNet

Method	**FID** $\downarrow$	**NFE** $\downarrow$
DDPM	3.17	1000
DDIM (50 steps)	4.67	50
FM (OT path)	**2.99**	**142**
FM (diff. path)	5.74	310

ImageNet 64×64; NFE = number of function evaluations

Takeaways

- OT paths are **strictly better** than Gaussian/diffusion paths
- Fewer NFE than DDPM with better FID
- Simulation-free training is competitive with diffusion

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Rectified Flow (Liu et al., ICLR 2023)

"Flow Straight and Fast" — Xingchao Liu, Chengyue Gong, Qiang Liu (UT Austin)

Core idea: Learn an ODE that follows *perfectly straight* lines from x_0 to x_1.

$$\min_v \ \mathbb{E}_{(x_0,x_1)\sim\pi} \int_0^1 \left\| (x_1 - x_0) - v(x_t, t) \right\|^2 dt$$

where $x_t = (1-t)x_0 + t\,x_1$ and π is an **arbitrary coupling** of p_0, p_1.

Key difference from FM

The **coupling** π matters! Independent coupling $\rightarrow$ curved marginal flows. Rectified Flow introduces **Reflow** to straighten them.

The Reflow Procedure

Problem: With independent coupling $\pi = p_0 \times p_1$, trajectories *cross*, making the marginal flow curved even if each conditional is straight.

Solution — Reflow:

1. Train $v_\theta^{(1)}$ with independent coupling
2. Generate **new coupled pairs**: $(\tilde{x}_0, \tilde{x}_1)$ where $\tilde{x}_1 = \text{ODESolve}(v_\theta^{(1)}; \tilde{x}_0)$
3. Retrain $v_\theta^{(2)}$ on these straighter pairs
4. Repeat

Theorem (Liu et al., 2022)

Reflow is non-increasing in expected convex transport cost. Trajectories converge to **non-crossing straight lines**.

Result after 2–3 reflow steps: single Euler step $\approx$ high-quality sample!

Rectified Flow: Results & Impact

Method	FID ↓	NFE
DDIM (10 steps)	13.4	10
DDIM (100 steps)	4.67	100
Rect. Flow (1 step)	8.5	**1**
Rect. Flow (10 step)	3.7	10
2-Rectified (1 step)	**4.8**	**1**

CIFAR-10 (32×32)

Impact on industry:

- **Stable Diffusion 3** uses Rectified Flow in latent space
- **FLUX** models (Black Forest Labs) are all Rectified Flow based
- Enables real-time generation with 1–4 steps

Also works for **image translation** (noise → data, domain A → domain B) with no modification.

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Stochastic Interpolants (Albergo et al., NeurIPS 2023)

"Stochastic Interpolants: A Unifying Framework for Flows and Diffusions" — Albergo, Boffi, Vanden-Eijnden (NYU Courant)

General stochastic interpolant:

$$I_t = \alpha(t)\, x_0 + \beta(t)\, x_1 + \gamma(t)\, z, \quad z \sim \mathcal{N}(0, I)$$

with boundary conditions: $\alpha(0) = 1, \beta(0) = 0$ and $\alpha(1) = 0, \beta(1) = 1$.

Special cases:

- $\gamma(t) = 0$: **Deterministic flow matching**
- $\alpha(t) = \sqrt{1-t}$, $\gamma(t) \propto \sqrt{t}$: **DDPM diffusion**
- Schrödinger bridges as minimizers

Training objective:

$$b^* = \arg\min_b \mathbb{E}\left\|\partial_t I_t - b(I_t, t)\right\|^2$$

A single least-squares regression unifies all methods.

Conditional Flow Matching with Minibatch OT (Tong et al., TMLR 2024)

"Improving and Generalizing Flow-Based Generative Models with Minibatch OT"
Tong, Fatras, Malkin, Huguet, Zhang, Rector-Brooks, Wolf, Bengio (Mila)

Key insight: The *source distribution* need not be Gaussian, and the *coupling* matters.

OT-CFM Objective

$$\mathcal{L}_{\text{OT-CFM}}(\theta) = \mathbb{E}_{t,\,(x_0,x_1)\sim\hat{\pi}_{\text{OT}}} \left\| v_\theta(x_t, t) - (x_1 - x_0) \right\|^2$$

where $\hat{\pi}_{\text{OT}}$ is a **minibatch optimal transport plan** (Sinkhorn or exact LP).

Why OT coupling?

- Pairs each x_0 with its *nearest* x_1 in the minibatch
- Produces straighter flows $\rightarrow$ fewer NFEs, more stable training
- Works with **any source distribution**, not just Gaussian
- Enables non-trivial source-to-target transport (e.g., single-cell RNA dynamics)

A Unified View: What Differs Between Methods

All flow matching methods share the same structure:

$$x_t = \text{interpolate}(x_0, x_1; t), \qquad u_t = \frac{d}{dt} x_t, \qquad \mathcal{L} = \mathbb{E}\left\| v_\theta(x_t, t) - u_t \right\|^2$$

Method	**Path**	**Coupling**	**Space**
FM (Lipman '22)	Linear OT	Independent	Pixel
Rectified Flow	Linear	Reflow	Pixel
OT-CFM (Tong '23)	Linear	Minibatch OT	Any
Stoch. Interp.	General α, β, γ	Independent	Any
Latent FM	Linear OT	Independent	Latent
Riemannian FM	Geodesic	Independent	Manifold

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Flow Matching in Latent Space (Dao et al., 2023)

"Flow Matching in Latent Space" — Dao, Phung, Nguyen, Tran (VinAI Research), arXiv 2307.08698

Motivation: Pixel-space FM is expensive for high-resolution images.

Latent Flow Matching (LFM)

1. Encode data: $z = \text{Enc}(x)$ using a pretrained VAE
2. Train flow matching in the **latent space** $z \in \mathbb{R}^d$ ($d \ll$ pixel dim)
3. Decode generated latents: $\hat{x} = \text{Dec}(\hat{z})$

Theoretical guarantee:

$$W_2(p_{\text{data}}, p_{\text{LFM}}) \leq W_2(p_{\text{data}}, p_{\text{FM,lat}}) + \text{reconstruction error}$$

Results: Competitive FID with Latent Diffusion Models (LDM) on CelebA-HQ, FFHQ, ImageNet — with faster convergence. This is effectively the LDM + FM combination that powers modern models.

Riemannian Flow Matching (Chen & Lipman, ICLR 2024)

"Flow Matching on General Geometries" — Ricky T. Q. Chen, Yaron Lipman (Meta AI), arXiv 2302.03660

Motivation: Many real-world data live on *manifolds*:

- Protein backbone angles live on **tori** $\mathbb{T}^n$
- Orientations live in **SO(3)**
- Molecular frames in **SE(3)** $= \mathrm{SO}(3) \ltimes \mathbb{R}^3$
- Point clouds on curved **meshes**

Riemannian conditional path: Replace linear interpolation with **geodesics**:

$$x_t = \exp_{x_0}\big(t \cdot \log_{x_0}(x_1)\big)$$

Target vector field: $u_t(x_t \mid x_1) = \dfrac{\log_{x_t}(x_1)}{1-t}$ (log-map generalization)

Simulation-free whenever geodesics are in closed form (sphere, torus, SO(3)).

SE(3) Flow Matching for Proteins

FrameFlow — Yim et al., arXiv 2310.05297 — **FoldFlow** — arXiv 2310.02391

Protein backbone = sequence of SE(3) frames (R_i, t_i) for each residue.

Flow on SE(3):

- Noise: uniform distribution on SE(3)
- Data: real protein backbone frames
- Path: geodesic interpolation on SE(3)
- Learn: $v_\theta(R_t, t_t, t)$

FoldFlow variants:

- FoldFlow-Base (independent coupling)
- FoldFlow-OT (Riemannian OT)
- FoldFlow-SFM (stochastic + OT)

Results vs. FrameDiff (SE(3) Diffusion)

- **5× fewer sampling steps**
- **2× better designability** (self-consistency TM-score)
- FrameFlow: **23× faster** than GENIE

Flow matching > diffusion for protein structure generation!

Discrete Flow Matching (Gat et al., NeurIPS 2024 Spotlight)

"Discrete Flow Matching" — Gat, Remez, Shaul, Kreuk, Chen, Synnaeve, Adi, Lipman (Meta AI)

Goal: Extend flow matching to **discrete token sequences** (language, code).

Discrete conditional path:

$$p_t(x \mid x_1) = \mathsf{Cat}\big(x;\ (1-t)\cdot \mathbf{1}[x = x_0] + t \cdot \mathbf{1}[x = x_1]\big)$$

Training: Cross-entropy at each timestep (like masked LM training):

$$\mathcal{L} = \mathbb{E}_{t, x_1, x_t \mid x_1}\left[-\log p_\theta(x_1 \mid x_t, t)\right]$$

Two prediction heads:

- x-**prediction**: predict clean token
- ϵ-**prediction**: predict noise token

Specific path schedulers (cosine vs. linear) matter significantly.

1.7B parameter model:

- HumanEval: 6.7% Pass@1
- MBPP: 6.7% Pass@1
- Significantly narrows autoregressive gap

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Flow Matching vs. Diffusion: Summary

	Diffusion (DDPM/Score SDE)	**Flow Matching**
Training	Simulation-free ✓	Simulation-free ✓
Training cost	Low	Low
Trajectory	Curved (Brownian)	Straight (linear)
Inference steps	100–1000 typical	10–50 typical
1-step generation	Poor	Good (after reflow)
Likelihood	Via SDE $\leftrightarrow$ ODE	Via ODE (exact)
Source distribution	Gaussian (fixed)	Any distribution
Manifold data	Requires adaptation	Native (Riem. FM)
Discrete data	MDLM, SEDD	Discrete FM

Why Straight Flows Matter

NFE (Number of Function Evaluations) at inference:

The numerical error of an ODE solver scales as:

$$\epsilon \propto h^p \cdot \int_0^1 \left\| \frac{d^p v}{dt^p} \right\| dt$$

where p is solver order, h is step size.

If trajectories are straight: $\frac{d^2 v}{dt^2} \approx 0$
$\Rightarrow$ single Euler step can be exact!

Trajectory curvature = wasted compute

- Gaussian diffusion paths curve around origin
- OT linear paths go directly source $\rightarrow$ target
- Reflow further removes crossing/curvature

Practical rule: minimize trajectory curvature via better coupling, not better solvers.

Real-World Impact: FLUX & Stable Diffusion 3

Stable Diffusion 3 (Esser et al., 2024)

- Applies **Rectified Flow** in the latent space of a pretrained VAE
- Uses a **Multimodal DiT** (MM-DiT) architecture with text + image tokens
- Outperforms DALL-E 3, Midjourney v6 on typography and compositional prompts

FLUX.1 (Black Forest Labs, 2024)

- Rectified Flow + hybrid attention architecture
- 12B parameter model
- State-of-the-art text-to-image at time of release
- Open-source (FLUX.1-schnell, FLUX.1-dev)

Flow matching is now the dominant paradigm for text-to-image.

Open Problems & Future Directions

Theoretical gaps:

- Optimal coupling beyond OT (Schrödinger bridges)
- Convergence rates for discrete FM
- Generalizing reflow guarantees
- Tighter likelihood bounds in latent space

Algorithmic improvements:

- Distillation without reflow (consistency FM)
- Learned noise schedules / path shapes
- Better minibatch OT approximations

Applications:

- Protein design (already strong)
- Drug/molecule generation
- Video generation (temporal FM)
- Audio synthesis
- Robotics & planning
- Scientific simulation (weather, PDEs)

Discrete FM for LLMs:
Can non-autoregressive transformers close the gap?

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Summary

Flow Matching in one equation

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\, x_0,\, x_1} \left\| v_\theta\big((1-t)x_0 + t\, x_1,\ t\big) - (x_1 - x_0) \right\|^2$$

Sample noise x_0, data x_1, time t. Interpolate. Regress. Done.

Key takeaways:

1. **FM = CNF training without ODE simulation** — just regression onto conditional vector fields
2. **Straighter flows = fewer NFEs** — OT/reflow coupling is the key
3. **Unified framework** — diffusion, flows, Schrödinger bridges are all special cases
4. **Extensible** — latent space, manifolds, discrete tokens all work
5. **Industrial impact** — Stable Diffusion 3, FLUX, protein design

Key References

1. **Lipman et al. (2022)**. Flow Matching for Generative Modeling. *ICLR 2023*. arXiv:2210.02747
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Thank you!

Discussion

- Which coupling strategy is most practically useful?
- Can discrete FM replace autoregressive models for language?
- What are the limits of Riemannian FM for complex geometries?

Appendix: Proof Sketch — CFM = FM Gradients

Claim: $\nabla_\theta \mathcal{L}_{\text{FM}} = \nabla_\theta \mathcal{L}_{\text{CFM}}$

Key identity (marginalization):

$$u_t(x) = \int u_t(x \mid x_1)\, \frac{p_t(x \mid x_1)\, q(x_1)}{p_t(x)}\, dx_1$$

Expanding $\mathcal{L}_{\text{FM}}$:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}\big[\|v_\theta\|^2\big] - 2\mathbb{E}\big[v_\theta^\top u_t\big] + \text{const}$$

By the tower property (law of total expectation):

$$\mathbb{E}\big[v_\theta^\top u_t(x)\big] = \mathbb{E}_{x_1}\mathbb{E}_{x_t|x_1}\big[v_\theta^\top u_t(x_t \mid x_1)\big]$$

Therefore $\mathcal{L}_{\text{FM}}$ and $\mathcal{L}_{\text{CFM}}$ differ only by terms independent of θ. $\square$

Appendix: Connection to Score Matching

Score-based diffusion trains: $s_\theta(x_t, t) \approx \nabla_x \log p_t(x_t)$

Flow matching with Gaussian path trains: $v_\theta(x_t, t) \approx u_t(x_t)$

Connection via the probability flow ODE (Song et al., 2021):

$$\frac{dx}{dt} = f(x,t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \quad \text{(PFODE)}$$

For the DDPM noise schedule, the FM vector field equals:

$$u_t(x) = f(x,t) - \tfrac{1}{2} g(t)^2 \nabla_x \log p_t(x)$$

Flow matching with Gaussian paths recovers the score function (up to known coefficients). FM is strictly more general.